

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0625

Berry Rd 138 kV
May 17, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0625. The Interconnection Customer proposes a 154.9 MW battery energy storage facility interconnecting at the Berry Rd 138 kV substation in the Pennsylvania area. The request was received on June 04, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 213 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 32 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$131,207,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Berry Rd / Cortland 138 KV	Walnut / Clear Springs 345 KV	227.5	189	74.1
Morovis 8801 / Bullmoose 138 KV	Walnut / West Gallatin Substation 345 KV	220.6	248	12.4
Newcomerstown / Morovis 8801 138 KV	W Gallatin / Walnut 345 KV	195.2	289	12.4
Lsbia / Cortland 138 KV Line	Walnut / Clear Springs 345 KV L	185.6	220	74.1
Wescosville / Donyville 138 KV	Magic Kingdom / Okawville 138 KV	184.8	213	8.1
Berry Rd / Florence 138 KV	Walnut / Clear Springs 345 KV	182.9	211	25.9
Newcomerstown Substation / Valley Home Tap 2 138 KV	W Gallatin / Walnut 345 KV	178.8	341	12.4
Florence Substation / W Hamln 138 KV	Walut / Clear Springs Substation 345 KV	177.7	222	25.9
Boyd's Mill Hydro / W Hamlin 138 KV	Walnut / Clear Springs 345 KV L	174.8	232	25.9
West Gallatin / Meredosia 138 KV	W Gallatin / Felida 345 KV L	174.5	261	5.8
E Arlington / Vly Home Tap 2 138 KV	Walnt / W Gallatin 345 KV	173.6	362	12.4
Wescosville / E Arlington 138 KV	Magi Kingdom / Okawville 138 KV	173.0	218	8.1
Boyd's Mill Hydro / Mears Corners 138 KV	Walnut / Clear Springs 345 KV L	170.9	246	25.9
Mears Corners / Walnut 138 KV	Clear Springs Substation / Walnut 345 KV	167.8	262	25.9
Dornyville / Edwardsport Sta 138 KV	Okawville / Magic Kingdom 138 KV	165.9	251	8.1
E Arlington / Landis Delivery No 2 138 KV	Edwardsport Station / Mount Olive Substation 138 KV	161.6	225	7.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Edwardsport Sta Substation / Mount Olive 138 KV L	Magic Kingdom / Okawville 138 KV	161.1	272	8.1
Landis Delivery No 2 / Okawville 138 KV	Mount Olive / Edwrdsport Station 138 KV line	160.7	232	7.1
Meredosia / Satsop 138 KV L	West Gallatin / Felida 345 KV line	159.3	250	5.1
Magic Kingdom / Okawville 138 KV	Mount Olive / Edwardsport Sta 138 KV	159.1	242	7.1
Lebia / Belding Substation 138 KV	Clear Springs / Walnut Substation 345 KV line	157.2	243	74.1
Felida 500-345 kV Transformer	Talenenergy Martins Crk Llc Harwood / Felida 345 KV	150.4	873	13.2
Tombstone 500 kV/345 kV XFMR	Tombstone / Marcus Hook Energy Lp 500 KV L	148.7	940	6.2
Talenenergy Martin Crk Llc Allentown / Deque 138 KV	Walnut 345/1kV Transformer Bank	144.9	169	54.7
Felida 345 kV/138 kV Autotransformer	Felida / West Gallatin 345 KV L	139.4	560	7.1
Mount Olive / Vndalia 138 KV	Magic Kingdom / Okawville 138 KV	133.2	269	5.7
Hills Chpel / Wallenpaupack 345 KV	Walnut / West Gallatin 345 KV	130.4	1,212	6.5
Walnut / Clear Springs 345 KV line	Wallenpaupack Substation 500-345 kV Transformer	129.3	1,101	8.2
Quarry / Riparia 138 KV	Los Angeles Hts / Van Wagner 138 KV	109.0	191	13.3
Clear Springs 500-345 kV TX	Berry Rd / Cortlad 138 KV Line	105.2	1,228	28.6
Rec Laprairie Substation / Sumner 138 KV	Van Wagner / Los Angeles Hts 138 KV line	104.6	233	13.3
Riparia / Rec Laprairie 138 KV	Los Angeles Hts / Van Wagner 138 KV	103.4	223	13.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Berry Rd / Cortland 138 KV	Rebuild line with higher-capacity conductor	\$1,234,000	100% (\$1,234,000)
Bullmoose / Morovis 8801 138 KV L	Rebuild line with higher-capacity conductor	\$1,736,000	100% (\$1,736,000)
Newcomerstown / Morovis 8801 138 KV	Rebuild line with higher-capacity conductor	\$4,011,000	100% (\$4,011,000)
Lsbia / Cortland 138 KV Line	Rebuild line with higher-capacity conductor	\$2,743,000	100% (\$2,743,000)
Wescosville / Donyville 138 KV	Rebuild line with higher-capacity conductor	\$2,058,000	100% (\$2,058,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Berry Rd / Florence 138 KV	Rebuild line with higher-capacity conductor	\$5,147,000	100% (\$5,147,000)
Newcomerstown Substation / Valley Home Tap 2 138 KV	Rebuild line with higher-capacity conductor	\$3,333,000	100% (\$3,333,000)
Florence Substation / W Hamln 138 KV	Rebuild line with higher-capacity conductor	\$11,332,000	100% (\$11,332,000)
Boyd's Mill Hydro / W Hamlin 138 KV	Rebuild line with higher-capacity conductor	\$2,254,000	100% (\$2,254,000)
W Gallatin / Meredosia 138 KV	Rebuild line with higher-capacity conductor	\$5,895,000	100% (\$5,895,000)
Vly Home Tap 2 / E Arlington 138 KV	Rebuild line with higher-capacity conductor	\$2,590,000	100% (\$2,590,000)
Wescosville / E Arlington 138 KV	Rebuild line with higher-capacity conductor	\$1,182,000	100% (\$1,182,000)
Boyd's Mill Hydro / Mears Corners 138 KV	Rebuild line with higher-capacity conductor	\$3,548,000	100% (\$3,548,000)
Mears Corners / Walnut 138 KV	Rebuild line with higher-capacity conductor	\$2,663,000	100% (\$2,663,000)
Dorneyville / Edwardsport Sta 138 KV	Rebuild line with higher-capacity conductor	\$5,008,000	100% (\$5,008,000)
Landis Delivery No 2 / East Arlington 138 KV	Rebuild line with higher-capacity conductor	\$5,325,000	100% (\$5,325,000)
Edwardsport Sta / Mount Olive 138 KV	Rebuild line with higher-capacity conductor	\$1,702,000	100% (\$1,702,000)
Landis Delivery No 2 / Okawville 138 KV	Rebuild line with higher-capacity conductor	\$5,008,000	100% (\$5,008,000)
Meredosia / Satsop 138 KV L	Rebuild line with higher-capacity conductor	\$14,292,000	100% (\$14,292,000)
Magic Kingdom / Okawville 138 KV L	Rebuild line with higher-capacity conductor	\$6,571,000	100% (\$6,571,000)
Belding Substation / Lesbia 138 KV L	Rebuild line with higher-capacity conductor	\$1,183,000	100% (\$1,183,000)
Felida 500/345 kV Auto	Replace transformer with higher-rated unit	\$25,200,000	11% (\$2,795,000)
Tombstone 500/345 kV Transformer	Replace transformer with higher-rated unit	\$23,170,000	3% (\$756,000)
Talenenergy Martin Crk Llc Allentown / Dequene 138 KV	Rebuild line with higher-capacity conductor	\$1,233,000	100% (\$1,233,000)
Felida 345/138kV Transformer	Replace transformer with higher-rated unit	\$21,008,000	100% (\$21,008,000)
Mount Olive / Vndalia 138 KV	Rebuild line with higher-capacity conductor	\$5,976,000	100% (\$5,976,000)
Hills Chapel / Wallenpaupack 345 KV	Rebuild line with higher-capacity conductor	\$57,939,000	3% (\$1,682,000)
Walnut / Clear Springs 345 KV line	Rebuild line with higher-capacity conductor	\$27,478,000	4% (\$1,232,000)
Quarry / Riparia 138 KV	Reconductor line	\$455,000	100% (\$455,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clear Springs 500/345 kV TX	Replace transformer with higher-rated unit	\$32,235,000	13% (\$4,120,000)
Rec Laprairie Substation / Sumner 138 KV	Reconductor line	\$2,679,000	100% (\$2,679,000)
Rec Laprairie / Riparia 138 KV	Reconductor line	\$456,000	100% (\$456,000)
	Total allocated to project		\$131,207,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.